

Introduction.

Visualization, Detection & Segmentation

1. GOAL

Introduction to medical imaging volumes: volume visualization in the 3 anatomical views (short axis, sagittal, coronal), structure of digital images.

Introduction to python environment: load and visualize data.

Introduction to detection and segmentation

2. MATERIALS:

Session slides, python code (Code.zip) and data volumes (CT.zip, MetadatabyAnnotation.xlsx) available at Campus Virtual.

3. SESSION EXERCICES

Exercise 1. Visualization of 3D Volumes. Download the materials to a local folder, launch Spyder 5 from Desktop and open S1_Visualization.py. Set the variables SessionPyFolder, SessionDataFolder to your local path. Otherwise, use Slicer3D available at <https://www.slicer.org>.

- a) Use the interactive visualization routines to browse through the volumes. What are intensity values for the lungs, soft tissue, bones and pulmonary lesion?
- b) What SA cuts show a lesion? How many lesions does LIDC-IDRI-0003 have?
- c) Install the Slicer3D software available at <https://www.slicer.org> and visualize lesions of the LIDC-IDRI-0003 case.

Exercise 2. Generate ROIs

- a) Use the mask to generate the bounding boxes. You can use Pyradiomics' Imageoperations' checkMask(). Install it using: pip install pyradiomics==3.0.1.
- b) Use the affine matrix to convert the bounding box coordinates of the metadata file to voxel coordinates. More information about it can be found at: https://nipy.org/nibabel/coordinate_systems.html

Exercise 3. Lesion Segmentation. Load and visualize one of the lesion ROIs.

- a) Set a value for thresholding, segment the lesion and visualize the segmentation
- b) Use the threshold set in a) to segment the remaining lesions. Discuss the quality of the results
- c) Define your own ROIs from the whole volume and segment the lesion using the threshold defined in a). Discuss the quality of the results